Research on Jumping Mechanism and Control Methods of Reconfigurable Robots

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***Abstract*—Reconfigurable wheeled mobile robots are promising for applications like resource exploration and disaster rescue. However, conventional wheeled robots face limitations when negotiating discontinuous terrain or confined spaces. This thesis investigates the modeling, control, and trajectory planning of a reconfigurable wheeled robot with a waist joint, enabling active jumping and robust transitions between four-wheel support and two-wheel balancing. An Euler-Lagrange dynamic model is derived and stabilized with linear controllers, including pole placement and LQR. To improve robustness, friction compensation and disturbance estimation are incorporated. A trajectory planning framework is formulated as an optimal control problem, solved using direct collocation. The proposed approach is validated through physical experiments, demonstrating stable balancing and reliable mode transitions.**

***Index Terms*—Reconfigurable wheeled robot, active jumping, Euler–Lagrange modeling, pole placement, LQR, friction compensation, disturbance observer, trajectory planning.**

I. Introduction

Reconfigurable wheeled mobile robots offer an efficient solution for a variety of tasks such as exploration, inspection, and disaster rescue, where mobility and versatility are crucial. Wheeled locomotion is particularly attractive due to its energy efficiency, simplicity, and advanced control methods. However, conventional wheeled robots often face limitations when navigating discontinuous terrain, including steps, gaps, or uneven surfaces, which are typically encountered in outdoor and real-world environments. These challenges arise because wheeled platforms require continuous ground contact and valid rolling paths, which are often disrupted by obstacles and terrain irregularities.

While various solutions exist to address terrain discontinuities, they can generally be categorized into three broad types: fully legged robots, hybrid wheel-leg systems, and wheeled platforms with auxiliary mechanisms. Fully legged systems, which provide high terrain adaptability, typically come with significant mechanical complexity and energy inefficiency. Hybrid wheel-leg robots, which attempt to combine the benefits of wheels and legs, are often similarly complex and energy-hungry. On the other hand, conventional wheeled robots excel in energy efficiency but lack the required mobility to handle obstacles such as steps and gaps. This introduces the need for a balanced solution that capitalizes on the advantages of wheeled locomotion while extending its adaptability to handle terrain discontinuities.

This thesis addresses this need by proposing a novel reconfigurable wheeled robot equipped with a waist joint. The waist joint allows for both configuration switching (e.g., from four-wheel support to two-wheel balancing) and active jumping capabilities. This unique approach enables the robot to extend its mobility beyond the constraints of pure wheeled locomotion, without the mechanical complexity or energy cost of legged systems. The primary focus of this research is to develop a system that can reliably and repeatably perform dynamic maneuvers such as jumping, while maintaining stability and efficiency during transitions between different motion modes, including driving, balancing, and jumping.

The proposed solution centers on a unified modeling, control, and trajectory planning framework that ensures the robot can handle various locomotion modes. A physically consistent Euler-Lagrange dynamic model is derived to represent both the wheeled locomotion and jumping phases. To ensure the robot can reliably balance and navigate discontinuous terrain, classical linear controllers such as pole placement and Linear Quadratic Regulator (LQR) are employed for balancing control. For dynamic behaviors such as jumping, a trajectory optimization framework is formulated as an optimal control problem and solved using direct collocation.

Additionally, the system is equipped with mechanisms to improve real-world robustness. Friction compensation and disturbance estimation, implemented via observer-based methods, ensure reliable performance even in the presence of

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external forces or sensor noise. The trajectory optimization framework focuses on the squat-to-take-off phase of jumping, which is critical for generating the required impulse for take-off. Planned trajectories are computed offline, validated in simulations, and then executed on the real robot with feedback tracking to ensure robust performance.

The effectiveness of the proposed approach is validated through a series of experimental tests. These experiments include tasks such as four-to-two-wheel transitions, two-wheel balancing control, speed tracking, disturbance rejection on uphill slopes, and jumping tasks (both long and high jumps). The experimental results demonstrate that the robot is capable of reliably performing dynamic maneuvers, including jumping, while maintaining stable locomotion and smooth transitions between different operational modes.

This thesis makes a significant contribution by introducing a reconfigurable wheeled robot that combines the energy efficiency of wheels with the added adaptability of jumping. This work bridges the gap between pure wheeled locomotion and legged systems, offering a unique solution to mobility challenges in complex, real-world environments. The proposed system not only shows the potential of reconfigurable robots for various applications but also sets the stage for further exploration of hybrid mobility systems in robotics.

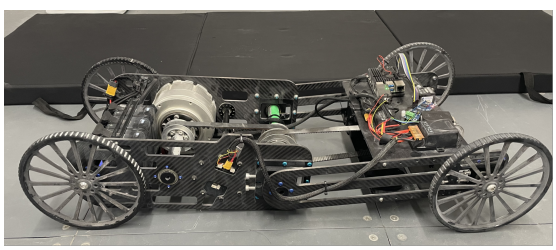

Fig. 1. The Reconfigurable Wheeled Robot.

II. SYSTEM DESCRIPTION

A. Mechanical Design

B. Hardware

To achieve a greater jumping height, the waist actuator must be capable of generating sufficiently large peak torque. Therefore, a Deep Robotics J1000P motor, with a peak torque of 140 Nm, was selected for the waist joint. The waist motor is driven by an Elmo Twitter driver, which communicates via CANopen and is connected to the onboard computer through a CAN-to-SPI interface board.

For the wheels, Unitree A1 motors were adopted, each of which provides built-in torque, velocity, and position control. An inertial measurement unit (IMU) is mounted on the lower part of the robot. The IMU roll angle and angular velocity are used as the pitch angle and pitch angular velocity of the lower rigid body in the subsequent model and control design. This choice is motivated by the fact that, whereas the pitch angle is limited to the range $[-\pi/2, \pi/2]$, the roll angle spans the full range $[-\pi, \pi]$, thereby enabling attitude measurement over arbitrary robot configurations.

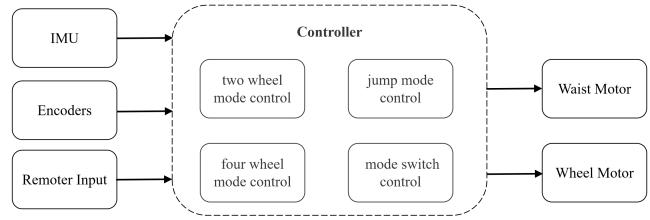

Fig. 2. Overview of the control architecture.

In addition, a LiDAR sensor is installed for obstacle detection and for triggering the jumping command. The main control unit is a Raspberry Pi 4B single-board computer equipped with a Broadcom BCM2711 CPU (1.5 GHz, 64-bit, quad-core, ARM Cortex-A72 architecture). To provide sufficient voltage and current for the waist motor, three batteries are connected in series as its dedicated power supply. The wheel motors and all other electronic components are powered by another separate battery. A complete list of hardware components is provided in Table 1.

TABLE I
HARDWARE COMPONENTS

Component	Name
$wheelmotor$	Unitree A1
$wheelmotorRS485toUSB$	emmm
$waistmotor$	Yunshenchu
$waistmotordriver$	Elmo
$CAN2SPI$	Wavesnow
$battery$	DJI
$OnboardComputer$	Raspi4B
IMU	Orientus
$IMURS232toUSB$	emmm
$Remoter$	Flysky
$RemoterSbustoUSB$	emmm
$Lidar$	360

C. Software

To ensure a sufficiently high real-time control frequency, the software system was developed in C++ based on the ROS 2 framework. The robot motion commands are provided through a remote controller. The control system consists of four-wheel driving control, two-wheel balancing locomotion control, jumping control, and mode-switching control. The overall control architecture is illustrated in Fig. 2.

III. SYSTEM MODEL

In this section, we introduce the modeling framework for the reconfigurable wheeled robot, which is essential for the subsequent control and trajectory planning tasks. The model consists of two primary components: the full-body model for jump trajectory planning and the simplified wheel-body model for balancing control. These models are used to describe the robot's dynamics during both the take-off and balancing phases, allowing us to design robust controllers and plan feasible trajectories.

A. Full-Body Model for Jump Trajectory Planning

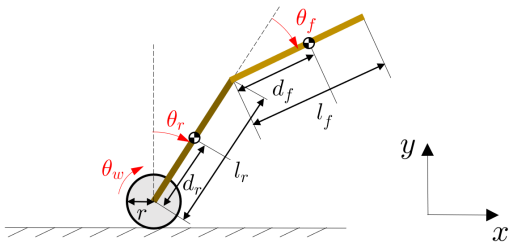

Fig. 3. The Full Dynamic Model.

The full-body model of the reconfigurable robot is developed using the Euler–Lagrange formulation. This model captures the dynamics of the robot during the take-off and flight phases. The robot consists of three rigid bodies: the rear wheel, the rear body, and the front body. The generalized coordinates for the system are defined as:

$$q = [\theta_r, \theta_b, \theta_w]^T \tag{1}$$

where θ_r is the rear body pitch relative to the vertical direction, θ_b is the waist relative pitch, and θ_w is the wheel rotation relative to the rear body. The generalized velocities are denoted as $q \doteq [\dot{\theta_r}, \dot{\theta_b}, \dot{\theta_w}]^T$.

The total kinetic energy T and potential energy V of the system are calculated for each rigid body:

$$T = \sum_{k \in \{r,f,w\}} \left(\frac{1}{2} m_k (\dot{X}_{k,1}^2 + \dot{X}_{k,2}^2) + \frac{1}{2} I_k \dot{X}_{k,3}^2 \right) \tag{2}$$

$$V = \sum_{k \in \{r,f,w\}} m_k g X_{k,2} \tag{3}$$

where m_k and I_k represent the mass and rotational inertia of each rigid body, respectively, and $X_{k,2}$ is the height of the center of mass (CoM) of each body. The Lagrangian $L = T - V$ is used to derive the equations of motion.

The full-body dynamics are written in the standard Euler–Lagrange form:

$$M(q)\ddot{q} + b(q, \dot{q}) = \tau_{\text{gen}} + f_{\text{nc}} \tag{4}$$

where $M(q)$ is the inertia matrix, $b(q, \dot{q})$ represents the Coriolis/centrifugal forces and gravity terms, τ_{gen} is the generalized actuator torque, and f_{nc} accounts for non-conservative forces such as friction.

B. Simplified Wheel-Body Model for Balancing Control

For real-time balancing control, a simplified model is used. This model treats the robot as an inverted pendulum on a wheel, which is a reduced-order system suitable for real-time control. The generalized coordinates are:

$$q = [\theta, \theta_w]^T \tag{5}$$

where θ is the body pitch angle relative to the vertical and θ_w is the wheel rotation relative to the body. The dynamics of the simplified model are described by:

$$M(q)\ddot{q} + b(q, \dot{q}) = \tau_{\text{gen}} \tag{6}$$

where the system matrix $M(q)$ is derived from the dynamics of the wheel and body, and the vector $b(q, \dot{q})$ contains the Coriolis forces and gravity terms. The generalized torque τ_{gen} is applied to the system through the wheel actuator.

C. State-Space Model for Planning and Control

The full-body model is used for jump trajectory planning, where the system's state includes both position and velocity. The simplified model, however, is used for balancing control and is linearized around the upright equilibrium point for easier controller design. The state-space representation for both models is as follows:

For the full-body model used in planning:

$$\dot{x} = f(x, u) \tag{7}$$

where $x = [q, \dot{q}]^T$ is the state vector and $u = [\tau_b, \tau_w]^T$ is the control input vector. The system dynamics are nonlinear, and the control input is related to the torques applied at the waist and wheels.

For the simplified wheel-body model used in control:

$$\dot{x} = Ax + Bu \tag{8}$$

where $x = [\theta, \dot{\theta}, \theta_w, \dot{\theta_w}]^T$ is the state vector, and $u = [\tau_w]$ is the control input, corresponding to the wheel torque. The matrices A and B are derived by linearizing the dynamics around the upright equilibrium point.

D. Model Integration and Hybrid Control

To bridge the full-body and simplified models, an equivalent center-of-mass (CoM) mapping is introduced. This allows the transition between different phases of the robot's motion, from take-off to landing, ensuring that both models can be used consistently in the planning and control phases. This integration of models provides a robust framework for both jump trajectory optimization and real-time balancing control.

E. Summary

In summary, this chapter presents the modeling framework for the reconfigurable wheeled robot. The full-body model provides a detailed representation of the robot's dynamics during jumping and take-off, while the simplified wheel-body model is used for balancing control. Both models are integrated into a unified framework that supports reliable transitions between motion phases, ensuring that the robot can perform dynamic maneuvers such as jumping and stable locomotion in real-world environments.

IV. Control and Planning Design

In this section, we present the comprehensive control and planning design for the reconfigurable wheeled robot, focusing on both the balancing control system and the jump trajectory planning. The goal is to develop a control system that allows the robot to perform stable locomotion in both wheeled configurations and dynamic maneuvers such as jumping. The design also incorporates trajectory optimization techniques to ensure that the robot can effectively handle terrain discontinuities while maintaining stability and efficiency.

A. Balancing Control Design

The balancing control system ensures that the robot remains stable in the upright position during locomotion. Given the nonlinear dynamics of the robot, particularly the coupling between the wheel and body, the system is modeled as an inverted pendulum. To stabilize the robot, we employ a state feedback control law designed using two approaches: pole placement and Linear Quadratic Regulator (LQR).

1) Pole Placement Control: Pole placement is applied to stabilize the system around the upright equilibrium. The state-space model of the simplified wheel-body system is linearized at the equilibrium point, and the control law is designed to place the system poles in the left half-plane to ensure stability. The control law is:

$$u = -Kx \tag{9}$$

where K is the state feedback gain matrix, and x is the state vector. The poles are selected based on the desired damping and response time, and the matrix K is calculated to achieve the desired system performance.

2) LQR Control: In addition to pole placement, Linear Quadratic Regulator (LQR) control is used as a conceptual baseline. LQR minimizes a cost function that balances the trade-off between control effort and deviation from the desired state. The cost function is given by:

$$J = \int_0^\infty (x^T Q x + u^T R u)\, dt \tag{10}$$

where Q and R are positive definite matrices that penalize state deviations and control efforts, respectively. The optimal control law is:

$$u = -K_{\text{LQR}} x \tag{11}$$

where K_{LQR} is the LQR gain matrix. The LQR controller is designed to ensure that the robot can maintain balance while minimizing the control effort. Though not implemented on the hardware, the LQR controller serves as an idealized performance baseline.

B. Friction Compensation

Drivetrain friction introduces a bias torque and can cause degraded low-speed behavior. To improve near-zero wheel-speed controllability, a simple smoothed Coulomb-like compensation term is added in feedforward. The following term is added to the commanded wheel torque:

$$\tau_{\text{fric}}(\dot{\theta}_w) = \tau_c \frac{\dot{\theta}_w}{|\dot{\theta}_w| + \epsilon} \tag{12}$$

where τ_c is the compensation amplitude and $\epsilon > 0$ is a smoothing factor that avoids a discontinuity at $\dot{\theta}_w = 0$. This heuristic compensation is intended to counteract Coulomb-like friction near zero velocity. The friction compensation parameters used in the experiments are:

TABLE II
Friction Compensation Parameters

Parameter	Value
τ_c	0.4 Nm
ϵ	0.1 rad/s

C. Swing-Up / Four-to-Two Wheel Transition

The swing-up and four-to-two wheel transition is an essential maneuver for the robot to move from a driving mode to a balanced mode. In this phase, the robot uses a reverse–forward torque profile to swing the body upwards and achieve a stable balancing configuration. This approach leverages actuator-limited capture with wheel-speed headroom.

The control logic for the transition uses a state feedback law that stabilizes the robot in the upright position while allowing dynamic torque profiles to achieve the transition. The system ensures a safe and repeatable transition to the two-wheel balancing mode.

D. Uphill Slope Disturbance Compensation

Real operation introduces persistent loads and biases, such as slope loads, drivetrain backlash, and parameter mismatches. To improve the robustness of the system during uphill operations, we implement a disturbance observer-based compensation. The observer estimates two slowly-varying disturbance channels: a pitch-bias term $d\theta$ that corrects the effective pitch state and an equivalent input torque disturbance $d\tau$ that is compensated in the wheel torque command.

The augmented-state disturbance observer is given by:

$$\dot{x}_a = A_a x_a + B_a u + L_o(y - C_a x_a) \tag{13}$$

where x_a is the augmented state including the disturbances, L_o is the observer gain, and y are the measured outputs from the IMU and encoders. The observer successfully compensates for disturbances, improving performance in uphill conditions.

E. Jump Trajectory Planning

Jumping is an essential feature for navigating terrain discontinuities, such as steps or gaps. The goal of the jump trajectory planning is to compute dynamically feasible trajectories that allow the robot to take off, fly, and land while adhering to actuator and physical constraints. We approach this as a nonlinear optimal control problem, solved using direct collocation techniques.

1) Squat-to-Take-Off Trajectory: The squat-to-take-off phase is critical for generating the necessary impulse to achieve jump take-off. The objective of the optimization is to maximize the vertical velocity at take-off, ensuring that the robot generates enough lift to overcome obstacles. The optimization problem is formulated as follows:

$$\text{maximize} \quad v_{\text{y, takeoff}} \quad \text{subject to:} \quad \tau_{\max} \geq \tau \geq -\tau_{\max}, \quad q_{\min} \leq q \leq q_{\max} \tag{14}$$

where $v_{\text{y, takeoff}}$ is the vertical velocity at take-off, τ is the actuator torque, and q represents the generalized coordinate. The optimization variables are the trajectory waypoints that define the robot's state at each time step, subject to the constraints of actuator torque limits, joint limits, and other physical feasibility conditions.

2) Direct Collocation and Optimization: To solve the trajectory optimization problem, we use the direct collocation method, which discretizes the continuous system dynamics into a set of discrete points. This allows us to apply optimization techniques to find the best trajectory while ensuring the feasibility of the dynamics at each point in time. The objective is to minimize the cost function:

$$\text{minimize} \quad \sum_{i=1}^{N} \left(\|\dot{x}_i - f(x_i, u_i)\|^2 + \lambda_u \|u_i\|^2 \right) \tag{15}$$

where x_i and u_i represent the state and control inputs at the i-th discretization point, and $f(x_i, u_i)$ is the system dynamics. The regularization term λ_u penalizes excessive control effort. This approach ensures that the planned trajectory is dynamically feasible while also minimizing energy consumption.

F. Integrated Control and Planning Framework

The control and planning modules are tightly integrated into a unified framework, where the trajectory planning output is fed into the control system. The controller tracks the planned trajectory and compensates for any discrepancies between the planned and actual states. The system operates in real-time, with feedback mechanisms in place to adjust the robot's behavior according to real-world conditions and disturbances.

The control system uses the feedback controllers designed in the balancing control section to maintain stability during dynamic maneuvers such as jumping. The robot's state is continuously monitored, and the control system adjusts the actuators accordingly to ensure that the robot follows the planned trajectory as closely as possible.

G. Simulation and Experimental Results

To validate the effectiveness of the proposed control and planning design, a series of simulations and real-robot experiments were conducted. The simulation environment was built using both MuJoCo and Webots to evaluate the robot's dynamic behavior, including its ability to perform stable locomotion, jump over obstacles, and transition between different motion modes. The simulations provided insights into the system's performance and allowed for the fine-tuning of control parameters.

1) Simulation Setup: In the simulation, the robot was subjected to various terrain conditions, including steps, gaps, and slopes, to test its ability to perform dynamic maneuvers such as jumping. The control system was validated by comparing the planned trajectories with the actual robot motion. The results demonstrated that the robot could successfully transition between different configurations (four-wheel support and two-wheel balancing) and perform jumps with minimal deviation from the planned path.

2) Experimental Setup: The proposed system was implemented on a physical prototype, and a series of experiments were conducted to test the robot's performance in real-world conditions. The robot successfully demonstrated four-to-two-wheel transitions, two-wheel balancing, and jump maneuvers, both high and long jumps. The experimental results confirmed that the robot was able to reliably perform dynamic maneuvers while maintaining stable locomotion.

3) Results and Discussion: The experimental results showed that the robot was capable of performing repeatable and robust dynamic behaviors. The jump take-off velocities were consistent with the planned trajectories, and the robot demonstrated stable balancing even after landing from jumps. Disturbances such as uphill slopes were effectively rejected by the disturbance observer, and the robot was able to recover to stable locomotion mode after each dynamic maneuver.

H. Summary

This chapter presented the control and planning design for the reconfigurable wheeled robot. The balancing control system used pole placement and LQR methods to ensure stable locomotion, while the jump trajectory planning system employed optimization-based methods to generate feasible jumping trajectories. The integrated system was evaluated through extensive simulations and real-robot experiments, demonstrating the robot's ability to perform reliable and dynamic maneuvers. The results show that the proposed control and planning design successfully enables the robot to handle discontinuous terrain while maintaining energy efficiency and stability.

V. Conclusion and Outlook

This thesis presents the design, modeling, control, and trajectory planning of a reconfigurable wheeled robot equipped with an active waist joint, which allows the robot to perform dynamic maneuvers such as jumping and transition smoothly between different locomotion modes. The proposed system

leverages the energy efficiency of wheeled locomotion while incorporating the adaptability of jumping to overcome terrain discontinuities, such as steps and gaps.

The key contributions of this work include the development of a unified modeling, control, and planning framework that supports stable locomotion, dynamic behavior execution, and reliable transitions between motion modes. The modeling approach, based on the Euler-Lagrange formulation, captures the essential dynamics during both the take-off and balancing phases, while the control system incorporates classical pole placement and LQR techniques for balancing and optimization-based trajectory planning for jumping.

Experimental validation through both simulation and real-robot experiments demonstrates the effectiveness of the proposed system. The results show that the robot is capable of reliably performing dynamic maneuvers, including four-to-two-wheel transitions, two-wheel balancing, and jumping, while maintaining stability and efficiency in real-world conditions. The disturbance observer effectively rejects external disturbances, and the robot demonstrates repeatable performance across trials, highlighting the robustness of the system.

In terms of future work, several directions can be explored to further enhance the capabilities of the reconfigurable wheeled robot. First, the integration of advanced machine learning techniques for adaptive control could improve the robot's ability to handle unforeseen environmental conditions. Second, the robot's hardware could be further optimized for better torque and speed performance, enabling faster transitions and more efficient jumps. Lastly, future research could focus on multi-robot coordination for large-scale exploration tasks, where multiple reconfigurable robots work together to cover larger areas or perform more complex tasks.

Overall, this work contributes to the field of reconfigurable robotics by providing a versatile, energy-efficient solution for navigating complex environments. The integration of wheeled locomotion with dynamic jumping offers a promising approach for robots to handle a wide range of tasks, from rescue missions to exploration, with high efficiency and adaptability.

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